SIH26123 · EDGE-AI DISTRIBUTED FLEET COORDINATION

If the control room dies, the aisle should not.

Three warehouse AMRs. No central dispatcher. Each robot broadcasts, auctions work, and yields locally — including when a human blocks the corridor.

Prototype · Hyderabad 3PL rack floor · measured vs stop-and-wait

Indian 3PL warehouses already have the failure mode.

Narrow aisles

1-wide rack corridors. Two AMRs cannot pass. A picker or carton blocks the only gap.

Central WMS / 5G

Every move waits on the cloud. Latency, cost, and a single outage stop the shift.

Mixed traffic

Humans, forklifts, robots. Obstacles are not in the map. They appear mid-pick.

Stop-and-wait

Both robots see a conflict and freeze. Deadlock. Picks do not leave the aisle.

Two common answers both fail on a live floor.

Central reservation

A fleet computer locks cells for everyone. Fast on a whiteboard. Dead when the link dies. Judges will ask: does Robot A know Robot B through a scheduler?

Stop-and-wait

Safe and dumb. On a 1-wide aisle both freeze. Our baseline never finishes 6 jobs in 800 ticks.

Every robot is a planner + a radio.

They negotiate like drivers at an unsignalized junction — not like clients of a control room.

1

Sense

Local range-2 obstacle + peer pose/path.

2

Broadcast

Direct peer packets. No scheduler in the middle.

3

Auction

Distance + battery + congestion + workload.

4

Yield

UBPA right-of-way. One steps into a turnout.

5

Replan

Live block → new path or RELEASE + re-auction.

The dashboard is a spectator. It never assigns work.

Robot 1
local planner

Robot 2
local planner

Robot 3
local planner

Peer broadcast · PEER_STATE / BID / YIELD_QUERY · not a dispatcher

Conflict from planned_path

Local negotiation

UBPA yield / replan

Judge question: does Robot A know what Robot B is doing directly, or through something else?

Answer: directly, through the peer broadcast.

UBPA: an explainable right-of-way, not a token ring.

U	**Urgency** Hot order / perishable pick scores higher.
B	**Battery** Low battery should not sit in a deadlock.
P	**Priority** Role / SLA weight, still local.
A	**Arrival** Who reached the conflict first.

Four local rules. Zero central locks.

Auction cost

distance + battery penalty + congestion + workload. Lowest valid bid wins. Invalid if battery cannot finish pick+drop.

Space-time A*

Each robot plans around peer paths it has heard. The world does not reserve cells for the fleet.

UBPA yield

On head-on: lower score steps into a turnout. Negotiation packets: YIELD_QUERY / ACK / NACK.

Stalemate

If both have stale data for 8 ticks, lower id yields. Then replan. If pick is unreachable, RELEASE and re-auction.

Hyderabad 3PL rack map, not an empty grid.

Layout

25×11 cells. 1-wide rack aisles. Two highways. Depot and charger. Turnout bays so a yield is physically possible.

Traffic mix

Mouse click = human / forklift / fallen carton. Robots sense range 2, broadcast the obstacle, replan.

Radio reality

Indoor 5G/Wi-Fi is lossy: at 40% packet loss the same 6/6 still finish, just later (137–196 ticks vs 93 clean) with 0 deadlocks. Sliders are live.

Why here

Indian 3PLs buy AMRs faster than they buy a second control room. Edge negotiation is the cheap resilience.

Same map. Same seeds. Baseline never finishes.

Do not say “X% faster.” There is no finish time to divide. Source: results/metrics.json

Experiment	Stop-and-wait	UBPA
E1 throughput	incomplete · 0/6 jobs · 2 deadlocks	93 ticks · 6/6 · 0 deadlocks
E2 head-on aisle	incomplete · 1/3 jobs · 2 deadlocks	50 ticks · 3/3 · 0 deadlocks
E3 aisle blocked t=16	incomplete · 3/6 jobs · 3 deadlocks	99 ticks · 6/6 · 0 deadlocks
8 robots · 16 tasks	incomplete · 1/16 jobs · 8 deadlocks · 1 stalls	327 ticks · 16/16 · 13 conflicts · 2 handoffs

How to read this

Stop-and-wait freezes on mutual occupancy, so distance is low and wait ≈ 1. UBPA travels farther because it finishes the jobs. Tick limit 800. comparable=false, so improvement % is n/a. Clean run finishes at 93 ticks.

Scale (probe, not headline): 8 robots / 16 tasks finish on 5/5 seeds at 327 ticks, 13 conflicts, 2 stall handoffs. Stop-and-wait finishes 0/5. 6 robots / 12 tasks finish 5/5 at 99 ticks; 10 robots / 20 tasks finish 5/5 at 180 ticks.

Attribution: the 5 fixes RC1-RC5 were each switched off. All off, 3-tick radio delay finishes 0/5 runs (577 intention conflicts); all off, 70% loss drops 5106 packets/run; RC5 off alone leaves the 8-robot run at 14/16 tasks with 37 deadlocks and no finish.

Published trade-off: with age-corrected peer reservations off (RC2) the 3-tick delay case still finishes but at 94 ticks with 3 intention conflicts, against 103 ticks and 0 conflicts shipped. We pay 9 ticks to remove those conflicts.

python scripts/run_demo.py → http://127.0.0.1:8765/

0:00	Three AMRs. No dispatcher. Point at the radio feed.
0:40	Auction: lowest valid cost. That is why a nearer robot took the pick.
1:10	Head-on in the 1-wide aisle. UBPA. One yields. Both continue.
1:50	BLOCK AISLE or click a cell. Replan or RELEASE + re-auction.
2:10	Drag packet-loss to 40%: same 6/6 finish, later (137–196 vs 93), 0 deadlocks.
2:25	Flip policy to stop-and-wait. Same map. They stall. That is the usefulness.

Honest path. We did not fake a warehouse pilot.

1

Now

Laptop sim. Same robot code. Spectator dashboard.

2

Next

One Raspberry Pi per robot. UDP multicast = this PeerBus.

3

Pilot

3 AMRs, one aisle, one human picker, one blocked carton.

4

Scale

Only after the 3-robot yield works on hardware.

What we do not claim

Real AMRs, production 5G mesh, on-robot GPU, 40-robot scale, or a signed 3PL deployment. Simulated: radio, lidar-range sense, battery drain, laptop clock as edge CPU proxy.

Robots that argue in the aisle
do not wait for a control room.

Who assigns tasks? Nobody. Each robot bids.

Who has right of way? UBPA, then robot id.

What if radio is stale? Stalemate timeout. Lower id yields.

python scripts/run_demo.py